SAAS BILLING & INVENTORY ERP SOLUTION

NexaBix System Prototype & Business Flow Architecture

A complete conceptual model, end-to-end user journeys, tenant isolation security, and high-fidelity screen prototypes for client review & sign-off.

ARCHITECTURE TYPE	CORE TARGET	DOCUMENT STATUS
Multi-Business SaaS (Cloud ERP)	**Fast POS, Billing & Inventory**	**Pending Client Sign-Off**

Notice for Client Evaluation: This document outlines the proposed end-to-end workflows, approval gating mechanisms, stock auto-deduction engines, and dashboard ergonomics prior to active engineering commencement.

1. Business Onboarding & Super-Admin Approval Pipeline

Strict multi-tenant security architecture ensuring unverified businesses cannot access or process records until explicit platform administrator authorization.

1. Self Registration — GSTIN, Store & Admin info → **2. Gated Lock** — IsApproved = false (Locked) → **3. Admin Review** — Super-Admin Audit Panel → **4. Active Tenant** — Full POS & ERP Access

1 Tenant Self-Registration & Account Creation

PUBLIC GATEWAY

The merchant signs up with Business Name, GSTIN, Trade Type, Currency, Address, and initial Admin credentials. The record is inserted into the `Businesses` table with an explicit security flag: `IsApproved = false`.

2 Security Middleware & Activation Lock Screen

ENFORCEMENT LAYER

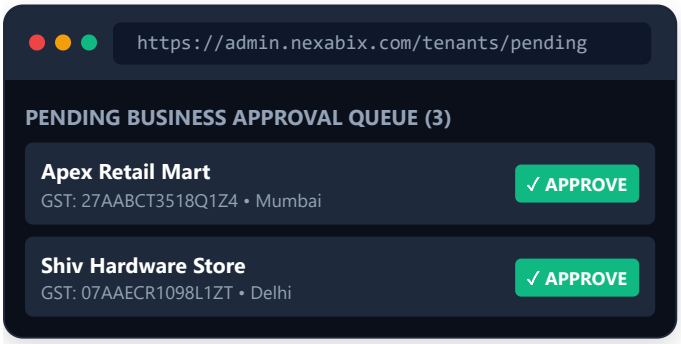
When the registered merchant logs in, a custom .NET Core Authorization Middleware evaluates the token's `BusinessId`. Since `IsApproved == false`, all operational routes (POS, Inventory, Reports) are intercepted and redirected to a clean, branded "Account Pending Approval" lock screen.

3 Super Admin Approval & Tenant Provisioning

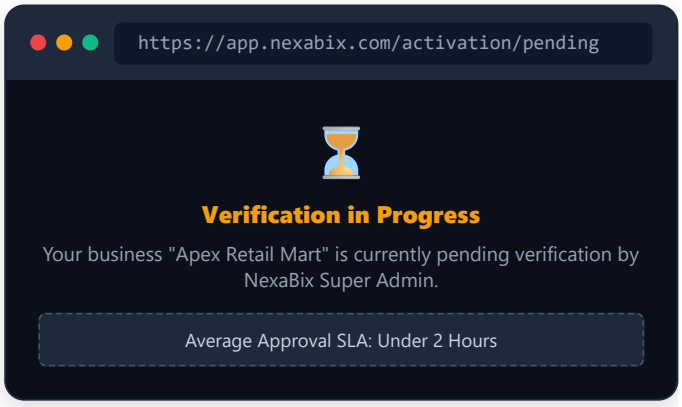
PLATFORM GOVERNANCE

Platform Super Admins access a dedicated Super-Admin Center. They review incoming businesses, inspect KYC/GST details, and click "Approve". `IsApproved` flips to `true`, instantly releasing all modules and sending a welcome notification.

Super Admin Approval Terminal



Merchant "Pending Activation" View

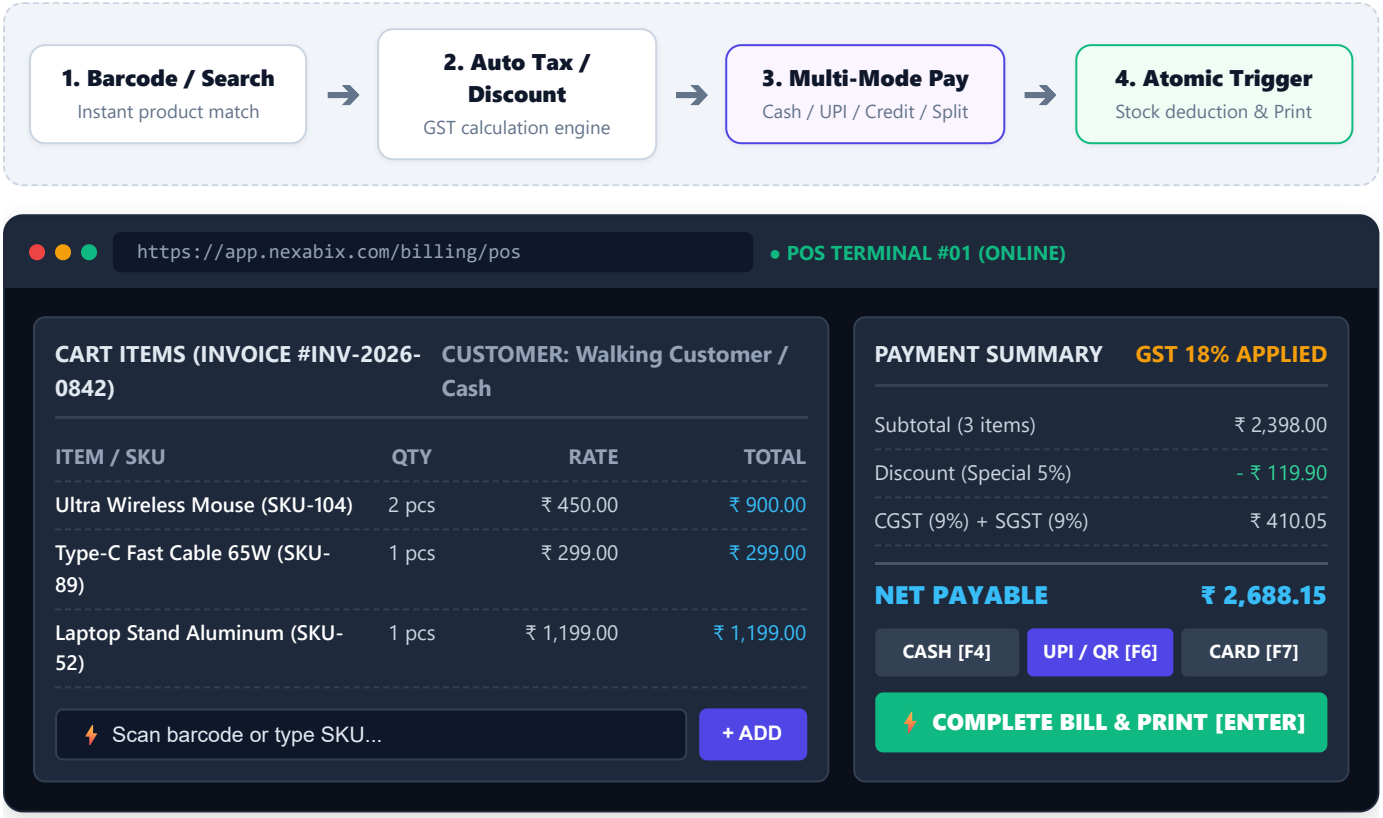


🔒 Multi-Tenant Data Isolation Principle

Every single SQL query across Products, Invoices, and Ledgers is enforced with an EF Core Global Query Filter: `builder.Entity<T>().HasQueryFilter(e => e.BusinessId == currentTenantId)`. No business can ever access, modify, or view another business's financial data.

2. Fast POS Billing & Automated Inventory Trigger Flow

A 3-second rapid billing sequence designed for high-throughput retail & wholesale counters with atomic stock deduction and ledger synchronization.



Real-Time Stock Deduction Engine

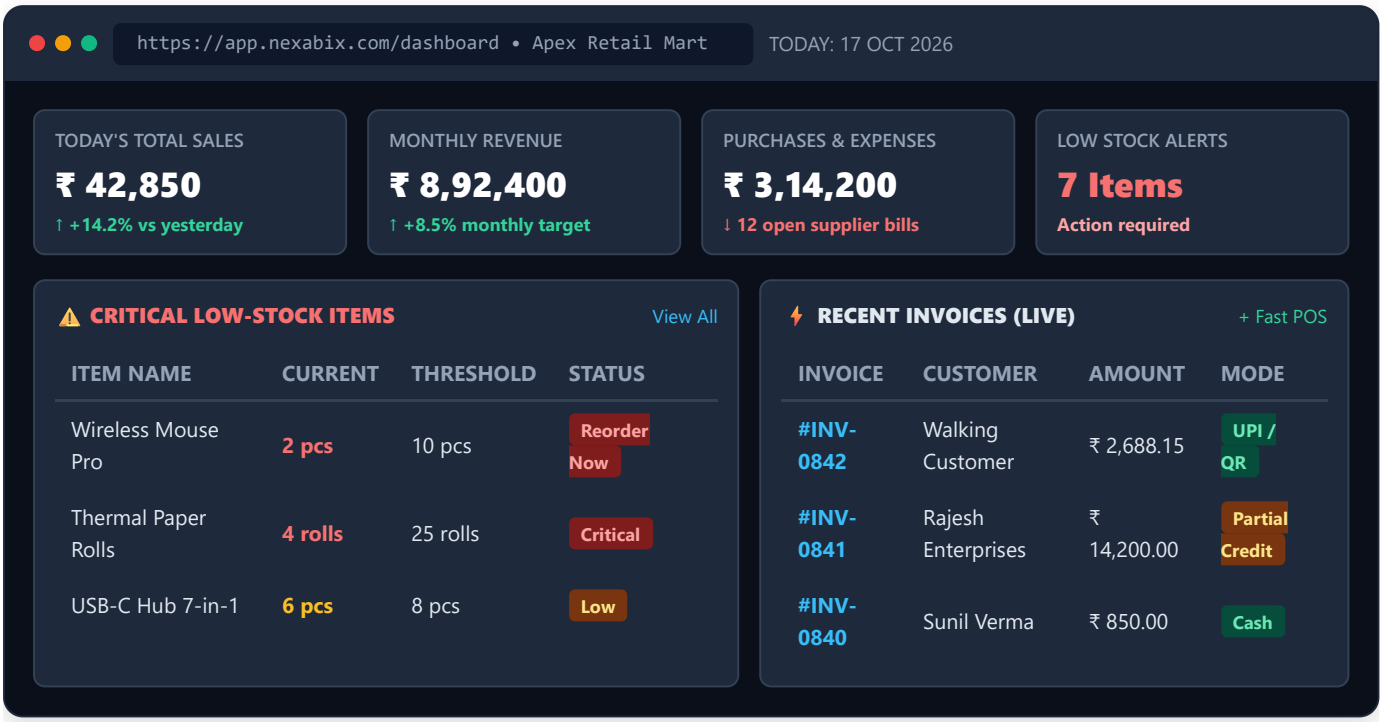
- **Atomic Database Transaction:** Stock quantity is decremented simultaneously with invoice generation inside a single `BeginTransactionAsync()` block.
- **Negative Stock Safeguard:** Configurable setting to prevent overselling or alert the cashier if on-hand stock is insufficient.
- **Auto Reversal:** If an invoice is voided or items returned, stock quantities are dynamically credited back to the warehouse.

Low-Stock Intelligent Alert System

- **Threshold Monitoring:** Every product features a tailored `LowStockThreshold`.
- **Live Flagging:** When `StockQuantity <= LowStockThreshold`, the item is instantly pushed to the Low Stock Priority Grid.
- **One-Click PO:** Direct generation of Purchase Orders to the linked Supplier with pre-calculated reorder quantities.

3. Executive Analytics Dashboard & Inventory Grid

Modern, high-visibility SaaS control deck inspired by Stripe and Vercel aesthetics for real-time operational insights.



4. Party Ledger & Dual Cashflow Engine

Unified customer credit/debit book and supplier payables ledger eliminating third-party accounting friction.

A

Customer Credit Ledger (Khata / Receivables)

Tracks unpaid invoices, partial cash collections, credit limits, and automatic WhatsApp payment reminder links with integrated UPI QR codes.

B

Supplier Payables & Purchase Reconciliation

Maintains supplier running balances against inward goods bills, tracking due dates to ensure timely vendor discounts and prevent duplicate bill payments.

5. Database Schema Blueprint & Multi-Tenant Isolation

High-performance relational structure in Microsoft SQL Server with strict tenant boundaries.

TABLE NAME	KEY ATTRIBUTES	ISOLATION & INDEXING	PURPOSE
Businesses	BusinessId, Name, GSTIN, Currency, **IsApproved**, Status	PK clustered, Unique GSTIN	Root tenant profile with verification flag
Users	UserId, BusinessId, Email, PasswordHash, Role (Admin/Cashier)	FK BusinessId, IX_Business_Email	Identity & role-based route permissions
Categories & Products	ProductId, BusinessId, SKU, Barcode, StockQty, LowStockThreshold	IX_Business_Barcode, IX_Business_SKU	Stock ledger, pricing tiers & threshold alarms
Parties	PartyId, BusinessId, Name, Phone, Type (Customer/Supplier), Balance	IX_Business_Phone	Unified customer & supplier ledger book
Invoices & Items	InvoiceId, BusinessId, InvoiceNo, Subtotal, Tax, Net, PaymentStatus	IX_Business_InvoiceNo, IX_Date	Master-detail sales billing & tax audit
Payments & Txns	TxnId, BusinessId, InvoiceId, Amount, Mode (Cash/UPI/Bank), Date	IX_Business_Date	Daybook, cashflow reconciliation & audit

6. Proposed Phased Implementation Roadmap

Structured rollout plan ensuring complete client validation before production release.

PHASE 01 (CURRENT)
Blueprint & Approval
Client review of Prototype, Schema validation, and System Flow approval.

PHASE 02
Core Security & DB
.NET Core 8/9 setup, EF Core migrations, Admin approval middleware & auth.

PHASE 03
POS Billing Engine
3-second fast checkout, barcode scanner, tax engine, atomic stock trigger.

PHASE 04
Ledgers & Reports
Party credit/debit Khata, Daybook, GSTR summary, Mobile PWA optimization.

Client Feedback & Next Action
Once you review and approve this prototype & flow, we will begin active development of Phase 02.

Client Sign / Date

Lead Architect